



A fast computational framework for the design of solvent-based plastic recycling processes

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ABSTRACT

Multicomponent plastics cannot be processed using mechanical recycling technologies, hindering efforts to deal with plastic waste. Multicomponent plastics include multilayer plastic films, which are widely used for food and healthcare packaging. Multilayer films combine several layers (potentially dozens) of different polymers to protect products from external factors (e.g., oxygen, water, temperature, shock, and light). Solvent-based separation processes have emerged as a promising alternative to recycle these complex materials. For instance, the Solvent-Targeted Recovery and Precipitation (STRAPTM) process uses sequential solvent washes to selectively dissolve and separate constituent polymers from multicomponent plastic waste, including films. STRAPTM process design (separation sequence, type of solvents, and operating conditions) changes significantly depending on the design of the multilayer plastic film (e.g., number, types, and proportions of polymers). The ability to quickly quantify the economic and environmental benefits of diverse STRAPTM process designs is essential to accelerate the development of sustainable recycling processes and more recyclable multilayer film products. In this work, we present a fast computational framework that integrates molecular-scale models, process modeling, and techno-economic and life cycle analysis to quickly evaluate STRAPTM designs. The computational framework is general and can be used to study the processing of complex multilayer plastic waste streams that contain many layers. We highlight the different uses of the framework via targeted case studies.

1. Introduction

High plastic waste generation and low recycling rates are challenging problems that need to be addressed to mitigate the ongoing plastic waste crisis (Law et al., 2020). One of the main obstacles hindering plastic recycling is that several common plastic materials, such as multicomponent plastics, cannot be processed and repurposed via mechanical recycling (Garcia and Robertson, 2017; Vollmer et al., 2020). These materials are broadly used in packaging applications, such as food and healthcare (Geyer et al., 2017). The packaging sector comprises nearly 40% of all plastic produced worldwide, and projections suggest that this share could increase in the future (Stegmann et al., 2022). Proposed alternatives to address the plastic waste crisis include (i) developing new and scalable technologies to process difficult-to-recycle plastics and (ii) developing new plastics that are compostable or easier to recycle (by design) (Foundation et al., 2016).

Solvent-based technologies are a promising approach for processing multicomponent plastic waste (e.g., mixed post-consumer waste and multilayer film waste) because they can recover/separate their constituent components (e.g., polymers, ink, additives, or other contaminants) (Sánchez-Rivera et al., 2023; Yu et al., 2023; Chaudhari et al., 2023; Uekert et al., 2023). This method allows for the recycling of multilayer films, widely used in food packaging (Sánchez-Rivera et al., 2021; Wagner, 2016). These films, composed of several layers of different polymers, were pioneered by Howard and Lineburg in 1963 to replace other packaging materials economically and efficiently (Howard and Lineburg, 1963). The design combined distinct polymer layers to leverage their unique properties to protect food from diverse external factors such as oxygen, water, temperature, and light. Multilayer films revolutionized food packaging by significantly extending shelf life (key for mitigating food waste). Modern multilayer

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films are highly sophisticated, with a number of layers and polymers tailored to different applications (e.g., fresh meat, frozen food, cheese, coffee, and tea) (de Mello Soares et al., 2022). Multilayer films with several hundred individual layers are common (Wagner, 2016). Several solvent-based technologies for recycling multicomponent plastics have been investigated in industry and academia (Polystyvert, 2023; Mäurer et al., 2021; Creasolv, 2016; Walker et al., 2020; Phan et al., 2023; Xu et al., 2025). The fundamental principles behind most of these processes are similar, but they differ in the types of feedstocks, products, process conditions, stage of development, and economic and environmental performance (Chaudhari et al., 2023; Li et al., 2022; Uekert et al., 2023; Sánchez-Rivera et al., 2023; Yu et al., 2023; Maga et al., 2019; del Carmen Munguía-López et al., 2023).

This work focuses on a solvent-based approach known as the Solvent-Targeted Recovery and Precipitation (STRAPTM) process. The STRAPTM process uses sequential solvent washes to selectively dissolve and separate different components of multilayer plastic films (Sánchez-Rivera et al., 2021; Walker et al., 2020; Sánchez-Rivera et al., 2023, 2024). The solvents used need to be selected appropriately to dissolve only a target component in each step. Therefore, the selected solvent must have a high solubility for the target component and a low solubility for the other components of the film (Zhou et al., 2021, 2023). Identifying appropriate solvents and operating conditions (e.g., temperatures) to dissolve only the target component without dissolving the other components is a non-trivial task. For the STRAPTM process, molecular-scale thermodynamic models have been employed to predict temperature-dependent polymer solubilities. As part of these efforts, Zhou et al. (2023) recently reported a computational workflow that performs large-scale temperature-dependent polymer solubility predictions. This approach also reports a computational solubility database for eight common polymers and more than a thousand solvents at multiple temperatures. To validate such predictions, they measured selected solubilities experimentally. The database has enabled the creation of a computational tool that automates solvent selections for solvent-based processes.

Recycling processes for multicomponent plastics are difficult to design because the number of constituent polymers and contaminants strongly influences their configuration and operating conditions (e.g., solvent types, separation sequence, and dissolution temperatures). Previous process models for solvent-based methods have been reported for specific types of multicomponent plastics. For instance, designs have been reported for a multilayer film composed of four polymers (Sánchez-Rivera et al., 2021), a multilayer film made of three polymers and ink (Sánchez-Rivera et al., 2023), and polypropylene face masks (Yu et al., 2023). These approaches have also conducted economic and environmental analyses of the processes to demonstrate that solvent-based technologies are competitive to virgin resin production. While these efforts have helped estimate the economic and environmental feasibility of processes using specific waste feedstocks and identify process design bottlenecks, they have also revealed a need for general frameworks that can target different types of multilayer films and inform their design. Recent work has also highlighted the importance of considering economic and environmental impacts when selecting solvents for different polymers in solvent-based processes (Zhou et al., 2023; Ikegwu et al., 2024). They also demonstrated that experimentally selecting solvents and operating temperatures is challenging due to the wide range of possibilities originating from the large number of polymer components and available solvent space. Additionally, if the process is designed only considering solubility constraints, it could be economically and environmentally infeasible (del Carmen Munguía-López et al., 2023).

There is a need to develop computational design frameworks to quickly evaluate different separation process designs (including varying separation sequences, solvents, and process conditions). A given process design may include the separation sequence or order (which component is dissolved first), the solvents that enable this sequence, and the

required process conditions, such as temperatures. Such insights can also help improve product design by identifying those polymers or components that lead to higher recycling costs and environmental impacts. Packaging designs often only consider economics and desirable properties (e.g., oxygen, water, and temperature barriers). However, the current plastic pollution crisis demands the recyclability of all plastic packaging to be considered in the product design directly (Foundation et al., 2016).

In this work, we present a fast computational framework that integrates molecular-scale models, process modeling, techno-economic analysis (TEA), and life cycle assessment (LCA) to identify sustainable configurations for solvent-based processes. The framework helps estimate the economic and environmental impacts of diverse process design scenarios, including different separation sequences, solvents, and process operating conditions. We also provide guidelines for product design by identifying multilayer film designs that are easier to recycle or have a lower recycling impact. The computational framework developed in this work is general and can be used for complex multilayer films or multicomponent plastic waste streams.

2. Methodology

2.1. STRAPTM process description and assumptions

The STRAPTM process uses selective dissolution and precipitation to recover the constituent components of multilayer films (Sánchez-Rivera et al., 2021, 2023, 2024). Fig. 1 presents a simplified schematic representation of the process considering a multilayer film with three components (in this case, each component represents only one polymer). Here, we can see that the recovery of each component involves a binary separation. That is, the recovery of the first target polymer (in this example, denoted by P_1 and depicted in yellow) is done by separating this component (via dissolution and precipitation) from the other components. In the same manner, the recovery of the second polymer (denoted by P_2 and depicted in orange) is done by separating this component from the third polymer (denoted by P_3 and depicted in blue). As mentioned previously, this is a simplified representation since common multilayer materials often have more than three components, and the STRAPTM process involves many more steps than the ones shown in Fig. 1. We also note that, even if polymer P_1 was targeted in the first binary separation step, we could also seek to target more than one polymer at a time (e.g., targeting both P_1 and P_2 , leaving P_3 undissolved). In addition, it should be highlighted that we could choose first to target P_2 or P_3 if we identify a solvent that can selectively dissolve P_2 or P_3 . This is possible because prior to the separation process via the STRAPTM process, the film is shredded, exposing more contact surface area with a selected solvent. As a result, the order of layers in the original film does not impact the separation order because it depends only on solubility constraints (Sánchez-Rivera et al., 2025; Zhou et al., 2023). Therefore, the feasible separation sequences could change depending on the solvents considered.

Fig. 2 presents the detailed process flow diagram for each binary separation. Here, we refer to each binary separation as a separation module and present the same example of a three-component material for a given sequence. In this example, each component comprises one polymer, and a couple of separation modules are required to recover all the polymers. The process steps for each separation module are as follows. The three-component multilayer film (represented as P_1 , P_2 , P_3), initially at temperature T_1 (typically room temperature), is heated to T_2 and mixed with a previously chosen solvent (that selectively dissolves only the target polymer, denoted as P_1). The elevated temperature, T_2 , is selected based on the predicted temperature that can dissolve only the target polymer with the chosen solvent (Zhou et al., 2023). Next, the solution is filtered to separate the undissolved solids (denoted as P_2 and P_3). Finally, the target polymer P_1 is precipitated by cooling to the temperature T_3 (typically room temperature), filtered, and recovered.

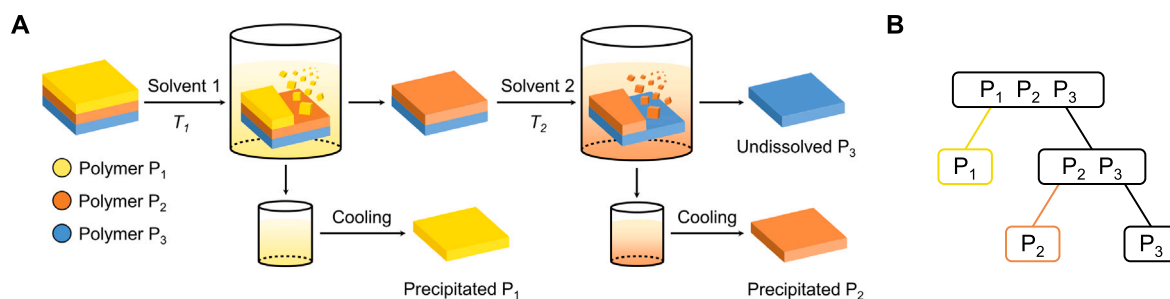


Fig. 1. Schematic representation of the STRAP™ process. (A) Simplified representation of the process considering a multilayer film with three components (in this case, each component represents only one polymer). (B) Separation process represented as a binary tree.

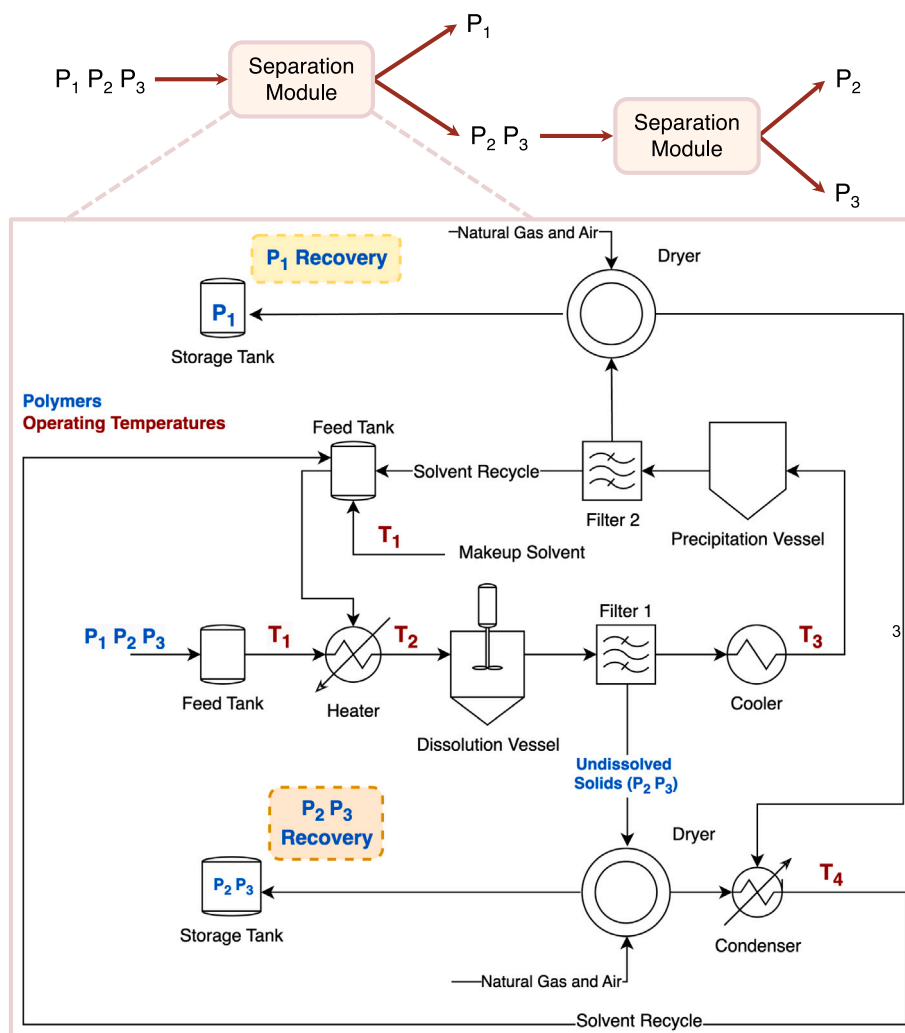


Fig. 2. Simplified process flow diagram of the STRAP™ process for one separation module.

Most of the solvent is also recovered and recycled. The solvent is recovered via filtration (Filter 2) and via a condenser that receives the vapor phase of the dryers. The process is repeated as many times as necessary to separate the remaining components (polymers P₂ and P₃), with the choice of solvent and operating temperatures potentially changing for each distinct component. From Fig. 2, we can see that separating an n-component multilayer film requires n–1 separation modules. Hence, for this example, we require two separation modules.

The separation module in Fig. 2 only considers temperature-controlled precipitation (also known as the cooling method) to recover

the target polymer from the solution. This step is usually achieved by reducing the temperature of the polymer solution to room temperature, where the dissolved target polymer becomes insoluble in the solvent. This method is economically and environmentally feasible and can be used for several polymer-solvent combinations (Sánchez-Rivera et al., 2021, 2023; Vollmer et al., 2020). Previous studies have also considered alternative precipitation techniques to recover the target polymer (Walker et al., 2020). Another method involves the addition of antisolvent. However, this method leads to higher energy requirements because a distillation unit is needed to separate the resulting mixture of

solvents (composed of the solvent used for dissolution and the antisolvent used for precipitation). Furthermore, antisolvent addition requires more solvents because recycling rates are lower (del Carmen Munguía-López et al., 2023). Evaporating the solvent directly from the polymer solution can also be used as an alternative process configuration to separate the target component. A disadvantage of this method is that its environmental impact is higher than the impact of temperature-controlled precipitation (Chaudhari et al., 2023). Therefore, it is critical to identify polymer-solvent combinations that allow precipitation via a temperature-controlled process at ambient pressure.

Previous studies have concluded that a high solvent recycling rate is needed to ensure the economic and environmental viability of solvent-based processes (Lehr et al., 2022; Schlummer et al., 2012; Munguía-López et al., 2024). Specifically, process costs and overall climate change impacts decrease with solvent recycling rates (Ügdüler et al., 2020). Moreover, it has been found that temperature-controlled precipitation can help achieve high recycling rates (>90%) compared to antisolvent-based precipitation (Yu et al., 2023; del Carmen Munguía-López et al., 2023). A sensitivity analysis for temperature-controlled precipitation considering different recycling rates identified that when recycling rates were equal or lower than 95%, the environmental impacts of the solvent-based process were higher than the impacts of producing the virgin resins from fossil sources (Sánchez-Rivera et al., 2023). Other studies have also agreed with these findings (Phan et al., 2023; Maga et al., 2019). In addition, previous work has reported that achieving such high recycling rates is feasible (Wagner and Schlummer, 2024; Creasolv, 2024). As such, in this work, we only consider solvents that can enable temperature-controlled precipitation and recycling rates higher than 95%, which is the lowest reported rate that allows for economic and environmental benefits. Some potential challenges related to achieving those recycling rates are that the energy required to do this can vary depending on the solvent. Therefore, certain solvents and sequences may have higher costs and emissions because of this (as we will observe in the Results and Discussion section).

2.2. Computational framework

We propose a computational framework that integrates molecular-scale models, process modeling, TEA, and LCA to provide fast economic and environmental analyses of different STRAP™ process design configurations and help identify sustainable designs. The process designs include assessing different separation sequences, solvents for every separation step, and process conditions (e.g., operating temperatures, recycling rate of solvents, and solid/liquid ratios). The computational framework follows a series of steps, summarized in Fig. 3 and described in the following subsections.

2.2.1. Separation sequences as binary trees

The first step consists of generating all the possible separation sequences as binary trees. Fig. 1B shows an example of the binary trees we use to represent the separation sequences for a multilayer film composed of 3 polymers. Here, we use the left branch to denote the dissolved and precipitated polymer(s) and the right branch to refer to the undissolved polymer(s). For this example, we can see that every separation step only dissolves one polymer. We refer to this as the dissolution number (D^n). Therefore, this example has $D^n = 1$. Also, this binary tree only shows one separation sequence: dissolving first P_1 and then P_2 . Changing the separation order of the sequence (for example, dissolving first P_2) provides all possibilities for the case of $D^n = 1$. Therefore, the permutation of the set of polymers (P_1 , P_2 , and P_3) results in 6 different sequences. To determine all the possible sequences, we also consider the binary trees where two polymers are dissolved in a certain step ($D^n = 2$). Considering the permutation of the polymers set, this results in another 6 different sequences. The maximum D^n for a n -component multilayer film is given by $n-1$. Therefore,

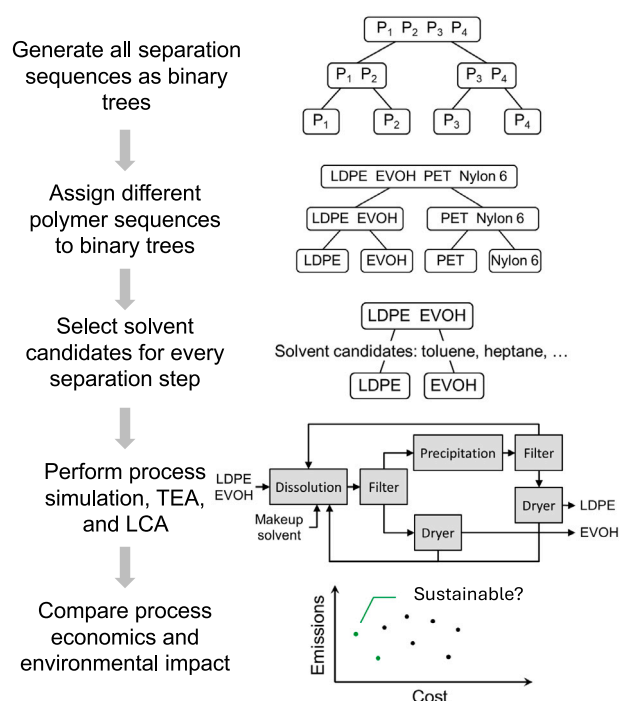


Fig. 3. General steps of the proposed computational framework.

the separation of a 3-polymer mixture has 12 possible sequences. Note that this combinatorial complexity increases rapidly with the number of polymers. For example, a 4-polymer mixture will have 120 sequences, and a 5-polymer mixture will have 1680 sequences.

As shown in Fig. 3, the next step is to assign different polymer sequences to the binary trees. For example, instead of having P_1 , P_2 , P_3 , and P_4 , now we will have low-density polyethylene (LDPE), which corresponds to P_1 ; ethylene vinyl alcohol (EVOH), which corresponds to P_2 ; polyethylene terephthalate (PET), which corresponds to P_3 ; and nylon 6 or N6, which corresponds to P_4 . This step is needed to select solvent candidates.

2.2.2. Solvent selection process

This step includes selecting the solvent candidates for the target polymer(s) in each separation step using molecular-scale models (Zhou et al., 2021). The selected solvents may vary depending on the separation sequence and the composition of the multicomponent material. For instance, for the multilayer film in Fig. 3, composed of LDPE, EVOH, PET, and nylon 6, different solvents may be chosen if LDPE or EVOH is dissolved first. In addition, if the multilayer film were composed of only LDPE, EVOH, and PET, the selected solvents might also differ.

The solvent selection is performed based on the previously reported solubility database and the solvent screening tool (Zhou et al., 2023). For the presented case studies, we consider a library of 60 common solvents selected from the solubility database. The criteria for selecting feasible solvents is that the target polymer(s) solubility should be greater than 10 wt% at an elevated temperature and less than 3 wt% at room temperature to enable dissolution and precipitation, respectively. Also, the solubility of the non-target polymer(s) should be lower than 3 wt%. These criteria were selected because they can lead to reasonable experimental yields, as previously reported (Zhou et al., 2023). The screening tool identifies all the solvents that can satisfy such criteria. If there are no feasible solvents for a given sequence, this sequence is discarded. Note that, at this point, solvent choice only depends on the temperature-dependent polymer solubilities. The economic or environmental impacts of solvent selection are considered in the following step (as discussed in the next section). Thus, the output of solvent selection

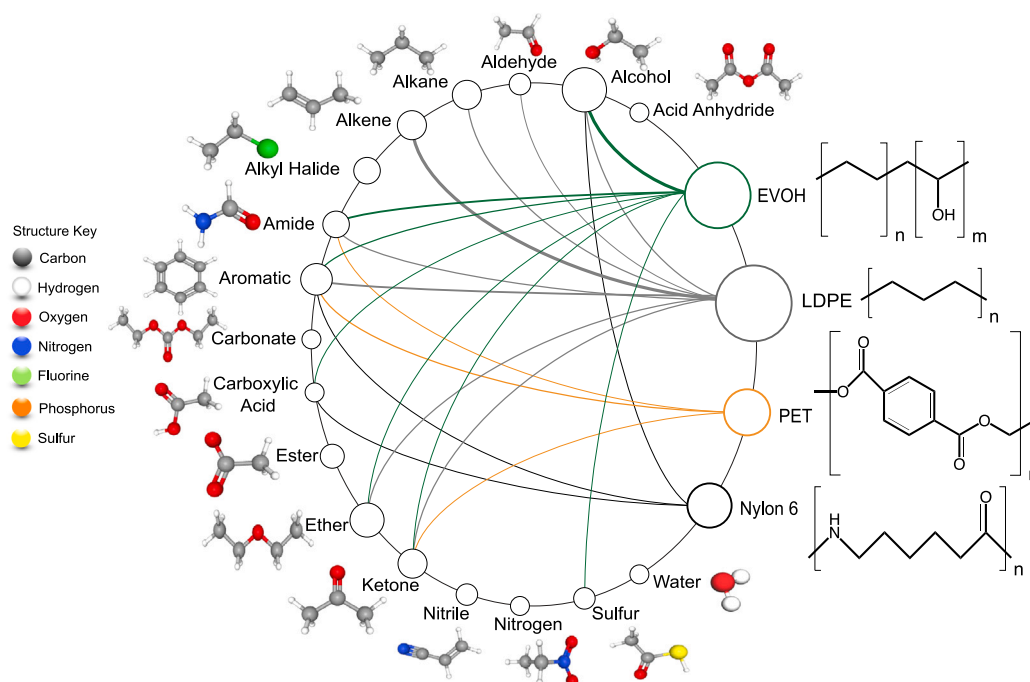


Fig. 4. Visualization of solvent candidates for EVOH (green lines), LDPE (gray lines), PET (orange lines), and nylon 6 (black lines); their chemical structures are drawn at right. Line widths and node sizes denote the number of solvents able to dissolve each polymer. Each solvent class has a representative chemical structure motif drawn next to it, depicting its primary functional group and a legend showing the type of atoms in each structure. For example, the Carboxylic Acid solvent class is characterized by the COOH group with the carbon, oxygen, and hydrogen atoms denoted in gray, red, and white, respectively.

only includes the solvent(s) for the target component in each separation step and the dissolution temperature.

The dissolution temperature for each solvent is determined following a previously reported approach (Zhou et al., 2023). In summary, the dissolution temperature is selected considering the solvent boiling point as follows: if the boiling point is greater than 120 °C, the selected temperature for dissolution is 120 °C. In contrast, if the boiling point is equal to or below 120 °C, the dissolution temperature is set to 1 °C below the solvent boiling point. The reason behind these limits is that the upper bound of 120 °C enables calculations of polymer solubility at a high temperature while limiting the selection of temperatures that may lead to thermal degradation of the target polymers (Cecon et al., 2022). Also, operating below the solvent's boiling point makes operating at ambient pressure conditions possible.

Fig. 4 presents a graphical representation of solvent candidates (from the entire solubility database, comprising 1000 solvents) that satisfy the criteria required to selectively dissolve each polymer in a multicomponent plastic containing LDPE, EVOH, PET, and nylon 6 polymers, assuming that the only polymers are present. Nylon 6 and PET have the fewest candidate solvents due to the polymer itself (e.g., not many solvents can dissolve PET) or to the specific combination of polymers in this example since finding solvents that selectively dissolve only one polymer can be challenging. On the contrary, LDPE and EVOH both have many solvents that can be used for their dissolution, which may indicate their ease of recovery. From this visualization (Fig. 4), we see that targeting either PET or nylon 6 polymers at the first dissolution step may be impossible or difficult, thereby reducing the separation design space. It is worth highlighting that Fig. 4 groups the 1000 solvents into 17 solvent categories (instead of each solvent) to facilitate their visualization, showing the diversity of the solvent database. However, the computational framework can provide the specific solvents for each process design as output.

2.2.3. Process simulation, TEA, and LCA

This step uses the collected inputs from the previous steps (feasible separation sequences, solvents, and dissolution temperature) to perform

the process simulation, TEA, and LCA of different STRAPTM process design configurations. We applied our framework to different case studies constituting various multicomponent plastic films containing known constituent polymers and mass fractions.

As indicated in previous studies, results have shown that a high solvent recovery (>95%) is required to ensure the economic and environmental feasibility of temperature-controlled precipitation (Yu et al., 2023; del Carmen Munguía-López et al., 2023). As shown in Fig. 2, after precipitation, most of the solvent is recovered after filtering the target polymer (using Filter 2). Then, the target polymer is sent to a dryer to recover the remaining solvent further. Similarly, the undissolved solids (P_2 and P_3) are also sent to a dryer to recover the remaining solvent in these components. The vapor phase of both dryers is then sent to a condenser to recover the solvent. Here, the solvent recovery rate is a function of the condenser temperature (represented by T_4 in Fig. 2). Therefore, we estimate the required temperature (T_4) in the condenser to achieve a solvent recovery rate of >99.99% in a computational pre-processing step (before applying the proposed computational framework). In this step, we performed the process simulation of a system consisting only of a dryer and a condenser. Several temperature points within a temperature domain (established based on previous work (Sánchez-Rivera et al., 2023; Yu et al., 2023)) are sampled and evaluated to achieve the required solvent recovery rate. In the scenario where a 99.99% recovery cannot be obtained, we estimate the temperature to recover >98% of solvent to maximize the economic and environmental benefits. The process model simulation is performed in the open-source package BioSTEAM. This Python-based process simulator has been previously validated against proprietary software (e.g., SuperPro Designer and Aspen Plus) using two published models for biorefineries as benchmarks (Cortes-Pena et al., 2020; Huang et al., 2016; Humbird et al., 2011).

We use the condenser temperatures previously estimated and perform the process model and TEA of all the feasible STRAPTM process design configurations in BioSTEAM. Here, mass and energy balances are used to determine the sizing and the energy required by the different process units. It is important to note that the process simulation

and TEA are performed for all the feasible sequences and modules (i.e., each binary separation) for the given multicomponent material. For example, given the material presented in Fig. 2, we simulate two separation modules for that given sequence since it has two binary separation steps. We repeat this step for all the feasible sequences. We use the minimum selling price (MSP) as the economic metric to compare different scenarios. The MSP is also known as the break-even price and is defined as the price per mass of product sold. We report the MSP as USD per kg of recovered polymer.

The TEA input parameters that are fixed across all separation configurations (including the cost of solvents) are presented in the **Supporting Information (SI)**. The SI also provides information on the estimated dissolution process conditions, including the dissolution and condenser temperature. Irrespective of the separation configuration, the polymer yield of the process, as well as the polymer inlet and precipitation temperatures, are fixed. These values are presented in the SI as well.

After performing the process simulation and TEA, we evaluate the environmental impact of each scenario using an LCA methodology. The LCA was performed using the open-source software openLCA v1.10.3 (Ciroth et al., 2020), the Environmental Footprint and AGRIBALYSE databases, and the Environmental footprint impact assessment method (Fazio et al., 2018; Colomb et al., 2015). We consider all the inputs to the process (electricity, steam, water) and estimate the climate change impact (CCI) for each scenario. Since there is limited data for solvents in the LCA databases and we are considering high recycling rates, we exclude the impact of producing the solvents used in each separation. The environmental impact of the process was analyzed considering a product perspective (Jeswani et al., 2021). From this perspective, the STRAPTM technology is seen as an alternative process to produce virgin-grade polymers. Therefore, the functional unit considered is the production of 1 kg of recovered polymer. As a result, the CCI is expressed in kg CO₂ eq. per kg of polymer. Evaluating other environmental impacts is out of the scope of this work. However, previous research has reported that a specific configuration of the STRAPTM process (using temperature-controlled precipitation) can have environmental benefits over virgin resins from fossil sources regarding climate change impact, energy use, water use, and toxicity (del Carmen Munguía-López et al., 2023). As presented in the Results section, the proposed work has also identified that various STRAPTM process designs can have a lower climate change impact than the production of virgin polymers. Therefore, these process designs could also provide environmental benefits for other metrics. Future work could include such an analysis.

2.2.4. Ranking of sustainable process designs

In this last step, we store the economic (MSP) and environmental (CCI) outputs of all feasible process design scenarios (including different sequences and solvents). Then, we compare and rank the economic and environmental impacts to identify sustainable process design configurations. Each of these configurations provides detailed process information such as solvent types and recycling rates, polymer/solvent ratios, and operating temperatures.

3. Results and discussion

We illustrate the applicability of our computational framework by addressing four case studies consisting of different feedstocks that can be processed using different STRAPTM designs. Table 1 presents a summary of the details of all case studies adopted in this work, the composition of each ML plastic as research objects, and the reason behind their selection. For Case Study 1, we analyze the economic and environmental impacts of recovering the components of a 4-polymer multilayer film commonly used for food packaging. For Case Study 2, we evaluate a more complicated multicomponent feedstock that consists of 7 polymers that can commonly be found in mixed plastic

Table 1

Summary of all case studies in this work.

Case study	Feedstock	Summary
1	4-polymer multilayer film: LDPE (60 wt%), EVOH (5 wt%), PET (25 wt%), Nylon 6 (10 wt%)	Suggests potential recycling processes for a common food packaging material, as a basic example usage of the computational framework.
2	7-polymer mixed plastic waste: LDPE (43 wt%), PET (26 wt%), PP (10 wt%), PC (6 wt%), EVOH (5 wt%), PVC (5 wt%), Nylon 6 (5 wt%)	Suggests recycling processes for a complex waste stream to demonstrate the applicability of the computational framework.
3	Two 3-polymer multilayer films: A: LDPE (75 wt%), PVC (5 wt%), & Nylon 6 (20 wt%). B: LDPE (75 wt%), EVOH (5 wt%), & Nylon 6 (20 wt%)	Compares the process design for two different films to study the effect of plastic waste compositions, and provides insights into the design of easy-to-recycle plastic products.
4	Different combinations within the 10 common polymers in the packaging industry	Evaluates the separation difficulty of plastic waste of different polymer components, and guides the design of plastic products that are suitable for solvent-based recycling.

waste streams. For Case Study 3, we compare the impact of changing the design of a 3-polymer multilayer film often used for food packaging applications. For Case Study 4, we analyze the separation feasibility in polymer mixtures considering various possible combinations between 10 common polymers. For the solvent selection, we consider a library with 60 common solvents based on our previously reported database (Zhou et al., 2023).

From previous research, we know the range of the required scale to achieve an MSP comparable to average market prices (Sánchez-Rivera et al., 2023). Therefore, the results presented in this work were obtained for a fixed plant capacity of 6400 tons per year. The feasibility of this plant capacity and the potential available feedstock can be assessed based on the information reported by other studies. For instance, a recent estimate reported that around 86% of the 44 million tons of plastic waste managed in the US were landfilled in 2019 (Milbrandt et al., 2022). Additionally, other studies have explored the feedstock availability for solvent-based plants of similar capacity. For instance, Chaudhari et al. (2023) studied the economic and environmental feasibility of a solvent-based process for PET packaging. They found that the feedstock demand for a plant of 8400 tons per year could be satisfied by a population of about 1.2 million people. Another example includes an economic analysis for a solvent-based plant to recover PP from face masks. This study reported that if 50% of the face masks discarded from the population of Wisconsin are available as feedstock, a plant capacity of 5292 tons per year could be installed (Yu et al., 2023).

3.1. Case Study 1

We applied the proposed computational framework to a multilayer film composed of LDPE (60 wt%), EVOH (5 wt%), PET (25 wt%), and nylon 6 or N6 (10 wt%). This 4-polymer multilayer film is commonly used for food packaging applications, such as ham (Wagner, 2016; CO, 2023).

We followed the steps in our framework as described and summarized in Fig. 3. Fig. 5 summarizes the process designs with the lowest economic and environmental impacts. We only present the feasible sequences based on the 60 common solvents in our database, denoted by S₁, S₂, S₃, and S₄. We include only the solvents that lead to the lowest environmental and economic impacts, which are the same for this case study. The impacts are given per separation step, and the

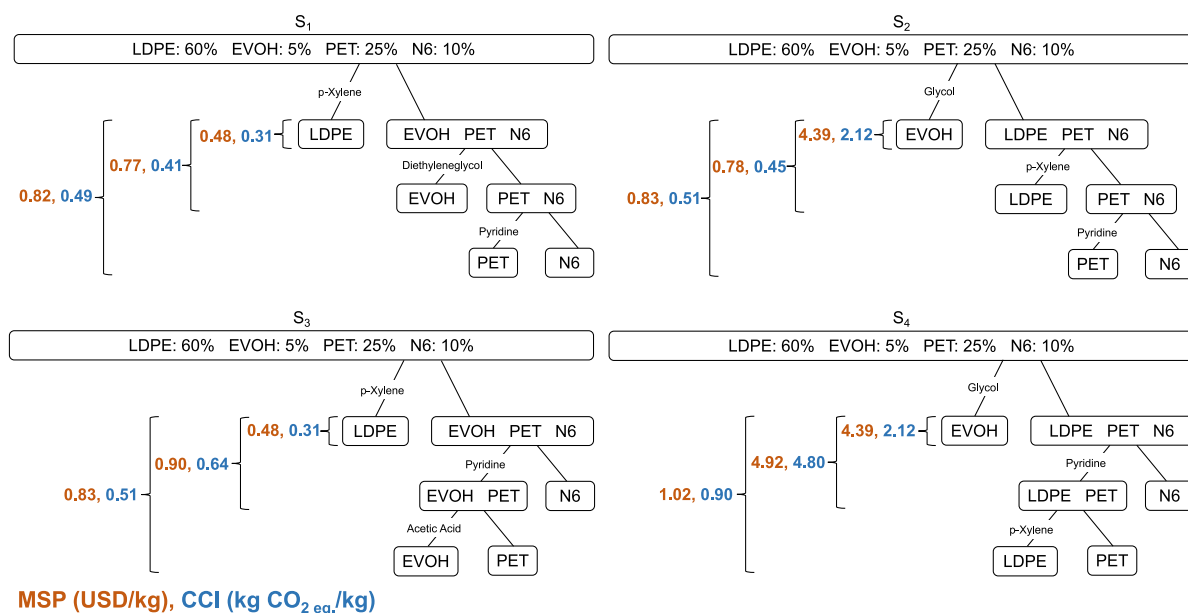


Fig. 5. Minimum selling price (MSP) and climate change impact (CCI) of the STRAP™ process configurations with the lowest impacts in Case Study 1. The separation sequences are denoted by S_1 – S_4 . Note that the impacts are given per separation step. For instance, for S_1 , the MSP when only LDPE is recovered is 0.48 USD/kg. Then, if LDPE and EVOH are recovered, the MSP is 0.77 USD/kg. Finally, if all the polymers are recovered, the MSP is 0.82 USD/kg.

solvent required for each step is also included. As mentioned in Section 2.2.1, for the binary tree representation, we use the left branch to denote the dissolved and precipitated polymer(s) and the right branch to refer to the undissolved polymer(s). Therefore, we can see that in the first step of S_1 , LDPE is dissolved using p-xylene as a solvent. The only product is LDPE with an MSP of 0.48 USD/kg and a CCI of 0.31 kg CO₂ eq./kg. Note that the process could end here if the only product of interest is LDPE. Following the next separation steps, EVOH is recovered in the second step. Since this step depends on the previous one, we estimate the MSP and CCI to be 0.77 USD/kg and 0.41 kg CO₂ eq./kg, considering the recovery of both LDPE and EVOH. Finally, in the last step, PET is dissolved, and N6 is recovered via filtration. This step includes the recovery of all polymers, and we refer to these MSP (0.82 USD/kg) and CCI (0.49 kg CO₂ eq./kg) as the total impacts.

As a benchmark, we also estimate the CCI of producing the four polymers from fossil sources and their average market prices. We find that the average market MSP of these polymers is 1.2–2.6 USD/kg. Furthermore, the multilayer film addressed in this study has a market price of 2.9 USD/kg (CO, 2023; Polymerscan, 2023). When we compare these costs with the total MSPs of the four sequences in Fig. 5, we observe that all the STRAP™ process configurations have lower costs. Regarding the environmental impact of virgin polymers, we estimate the CCI of the production from fossil sources of the multilayer film considered in this case study (LDPE, EVOH, PET, and N6). We find that 4.81 CO₂ eq./kg are generated. Fig. 5 shows that all the total CCI of the STRAP™ process configurations are lower than this value, supporting the claim that STRAP™ offers a better process for plastic production from an environmental standpoint (del Carmen Munguía-López et al., 2023).

If we compare the total impacts of each sequence, we can see that S_1 provides the lowest impacts, while S_2 and S_3 have similar impacts. However, if S_4 were chosen as the selected process design, we note that the cost and emissions would be 24% and 84% higher than S_1 , respectively. These differences greatly increase if we compare only the first step, which is the recovery of only one polymer. We compare S_1 and S_3 , which recover LDPE first, with S_2 and S_4 , which recover EVOH first. We find that separating only EVOH is 9 times more expensive and generates 7 times more emissions. When we compare the next separation step, which includes the recovery of two polymers, we find

that recovering EVOH and N6 via S_4 results in a 6 times higher MSP and 12 times higher CCI than using S_4 to recover LDPE and EVOH.

It should be highlighted that each step and sequence has different process parameters (e.g., temperatures, recycling rate, solvent/polymer ratio) that also contribute to the reported impacts. For instance, among all the separation steps, the selective dissolution of LDPE has the lowest MSP and CCI. This is partly due to the high weight percentage of LDPE in the multilayer plastic film (60%) and the high recycling rate (99.98%) of the solvent (p-xylene) used.

From these results, we conclude that recovering all polymers is the best option because it helps reduce the impacts. However, if the objective is to have only the first or second step of the process, S_3 and S_4 should be avoided because these sequences result in higher costs and impacts. In summary, while this analysis shows that a rule of thumb to follow to achieve a low MSP and CCI is to selectively dissolve components from the largest percentage composition to the lowest (taking advantage of the economics of scale), there are other factors and constraints to consider. For instance, although S_1 provides the overall best MSP and CCI by dissolving LDPE first, targeting PET next based on its percentage composition could lead to lower MSP and CCI. However, finding a solvent capable of selectively dissolving and precipitating PET from an EVOH-PET-N6 polymer mix is difficult. The same applies to N6. This shows that other factors related to the solvent also affect the MSP and CCI. For instance, this higher MSP and CCI could be attributed to the fact that only 99.95% of pyridine, which was used to recover N6 in S_3 , can be recycled at a much lower condenser temperature of 243.15 K. In contrast, 99.99% of diethylene glycol that was used to recover EVOH in S_1 could be recovered at a less energy-intensive condenser temperature of 323.15 K. Having identified the influence of the percentage composition, recycling rates, and condenser temperature on the MSP and CCI, other solvent effects like solvent dissolution energy could affect the CCI (Ikegwu et al., 2024).

3.2. Case Study 2

For this case study, we consider a 7-polymer feedstock commonly found in mixed plastic waste streams. We select the composition based on previous approaches: LDPE (43 wt%), PET (26 wt%), polypropylene (PP) (10 wt%), polycarbonate (PC) (6 wt%), EVOH (5 wt%), polyvinyl

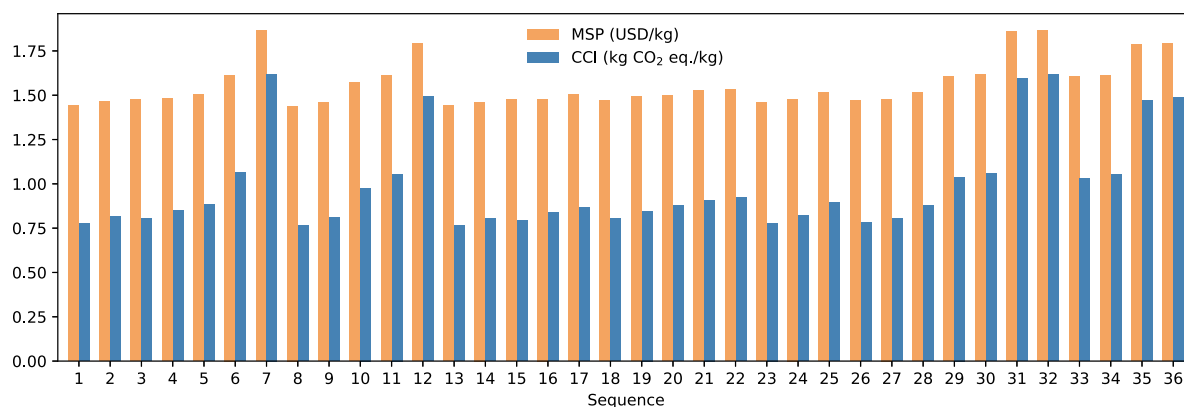


Fig. 6. Minimum selling price (MSP) and climate change impact (CCI) of the STRAP™ process configurations with the lowest impacts in Case Study 2. The presented impacts are the total cost and emissions of separating the seven polymers. Detailed information of each sequence (values, separation order, solvents) is available in the **Supporting Information**.

chloride (PVC) (5 wt%), and N6 (5 wt%) (Wagner, 2016; Sánchez-Rivera et al., 2025). Note that the main components are LDPE and PET, similar to Case Study 1. Here, we do not focus on a specific multilayer film because the goal is to analyze the feasibility and the impacts of recovering the polymers of a mixed waste stream that may contain different types of multilayer films or other multicomponent materials.

We followed the general steps for the proposed computational framework. First, when we evaluated all the possible sequences, we found that 36 different sequences were feasible for this mixture of polymers. In other words, we can recover the seven polymers using any of these 36 separation sequences, where the order of separation changes, as do the other process variables (e.g., solvents and temperatures).

Because different sets of solvents may be used to achieve a complete separation sequence, we select the process designs (i.e., the sequence with its required solvent set) of each of the 36 feasible sequences that lead to the lowest impacts. Fig. 6 presents the total impacts for all sequences: the cost and emissions of all the steps required to separate the seven polymers. Similar to Case Study 1, when we compare different process conditions (solvents and temperatures) for a given sequence, we find that the process parameters that lead to the lowest MSP also result in the lowest CCI. Therefore, the MSP and CCI presented in Fig. 6 represent each sequence's lowest economic and environmental impacts.

When we compare the MSP between the different sequences, we can see that the lowest MSP is given by Sequence 8 with 1.44 USD/kg. Here, the separation order is (1) EVOH, (2) PVC, (3) PP, (4) LDPE, (5) PC, (6) PET, and (7) N6. Then, Sequence 7 provides the highest MSP, resulting in 30% higher costs with 1.87 USD/kg. The separation order for this sequence is (1) EVOH, (2) PC, (3) PET, (4) N6, (5) PVC, (6) PP, and (7) LDPE. Contrary to Case Study 1, we can see that in this case, recovering the polymers with the highest composition (LDPE-43 wt% and PET-26 wt%) first does not lead to the lowest MSP. This is because the solvents used, which depend on the multilayer/mixed plastic composition, have a greater influence on the costs than the percentage composition; in other words, the solvents and conditions required to recover LDPE in a 4-component film change from the solvents and conditions to recover LDPE in a 7-component mixture to satisfy the selective dissolution criteria.

Similarly to the previous case studies, the same sequences that provide the lowest and highest MSP lead to the lowest and highest CCI. The difference between the minimum and maximum CCI is greater than the difference for the MSP. For Sequence 8, the CCI is 0.77 kg CO₂ eq./kg, while for Sequence 7, the CCI is 2.1 times more with 1.62. These results highlight the importance of estimating and comparing the impacts of all the feasible sequences.

3.3. Case Study 3

After demonstrating the relevance of evaluating different process designs and identifying the main impact drivers for the STRAP™ process, we examine the economic and environmental variations when the product design changes. Several companies are working to replace specific polymers in multilayer films used for food packaging to reduce environmental and human health hazards (Rodrigues et al., 2019; Godswill and Godspel, 2019). PVC is of particular concern and has already been replaced in some food packaging applications with EVOH (Amcor, 2024a,b; Napierska, 2024). Therefore, in this case study, we aim to analyze the impacts of changing such packaging designs on the STRAP™ recycling process. To do that, we compare a multilayer film composed of LDPE (75 wt%), PVC (5 wt%), and N6 (20 wt%) with a similar film where EVOH is used instead of PVC. These multilayer film designs are often used for cheese packaging (Amcor, 2024a; Wagner, 2016).

We refer to the LDPE/PVC/N6 film as Case Study 3A. Fig. 7 presents the process designs with the lowest economic and environmental impacts. Here, the impacts and solvents per separation step are shown. If we compare the sequences S₁–S₄, we can see relevant differences. For instance, when we compare S₁ and S₂, we can see that recovering only PVC results in a 10 times higher MSP and 5 times higher CCI. Similarly, S₃ and S₄ lead to 4 and 5 times higher costs and emissions than S₁ in the first separation steps. Note that in S₃ and S₄, the product is N6, but it is obtained via filtration while the dissolved polymers are LDPE and PVC (the polymers on the right branch of the tree as mentioned above).

Similar to the previous case studies, we observe that the impacts decrease if we consider the recovery of all components. However, there are important differences between the lowest and highest impacts. For instance, S₃ results in 33% and 94% higher MSP and CCI than S₁.

Fig. 8 presents the process designs with the lowest economic and environmental impacts for the LDPE/EVOH/N6 film, which we refer to as Case Study 3B and represents a re-designed film that replaces PVC with EVOH. We can see some similar trends to Case Study 3A. For instance, S₁ has the lower CCI and MSP. Then, for S₂, if only EVOH is recovered, the MSP and CCI are 11 and 7 times higher, respectively. S₃ and S₄ also result in higher impacts, but recovering all polymers reduces the impacts.

If we compare the recovery of only PVC (see Fig. 7) or EVOH (see Fig. 8) using S₂, we find that the MSP is similar, but the CCI is 15% higher for the EVOH separation. When considering the recovery of all polymers for both case studies, we observe no significant differences for S₁ and S₂. However, for S₃, the film containing EVOH results in 13% and 23% higher MSP and CCI, respectively, compared to the film

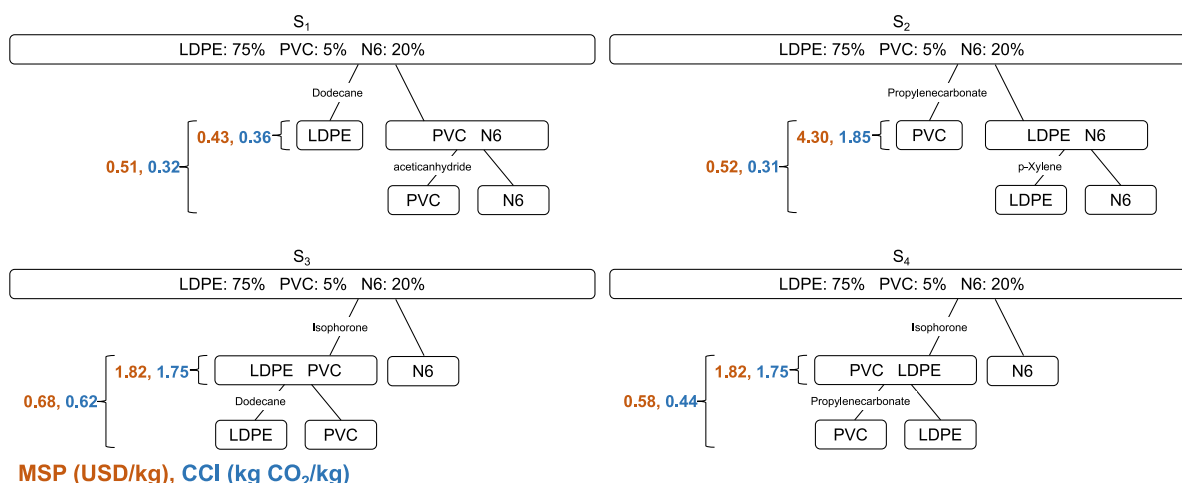


Fig. 7. Minimum selling price (MSP) and climate change impact (CCI) of the STRAP™ process configurations with the lowest impacts in Case Study 3A. The separation sequences are denoted by S_1 – S_4 . Note that the impacts are given per separation step. For instance, for S_1 , the MSP when only LDPE is recovered is 0.43 USD/kg. If all the polymers are recovered, the MSP is 0.51 USD/kg.

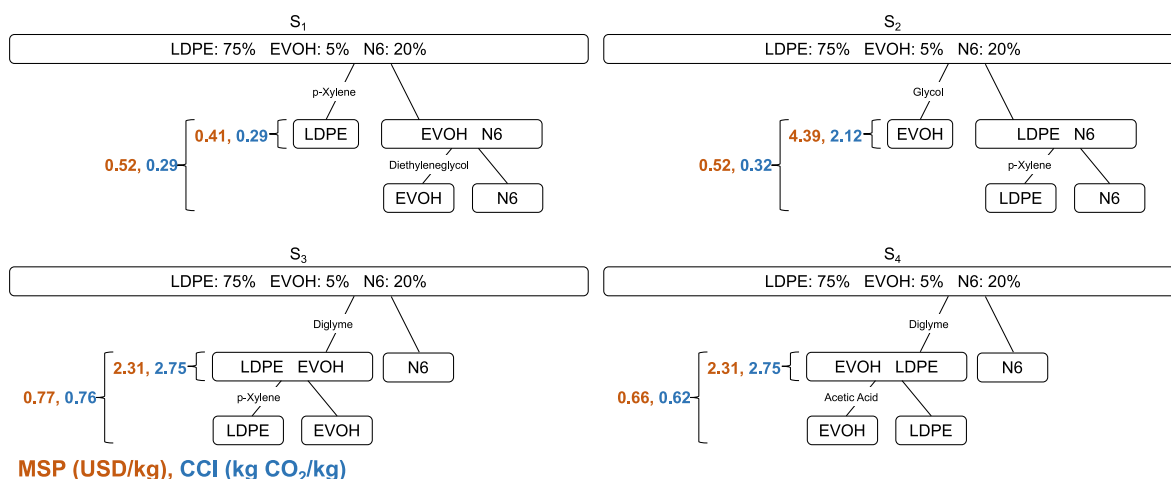


Fig. 8. Minimum selling price (MSP) and climate change impact (CCI) of the STRAP™ process configurations with the lowest impacts in Case Study 3B. The separation sequences are denoted by S_1 – S_4 . Note that the impacts are given per separation step. For instance, for S_1 , the MSP when only LDPE is recovered is 0.41 USD/kg. If all the polymers are recovered, the MSP is 0.52 USD/kg.

containing PVC. For S_4 , the increase in MSP is similar, but the CCI is now 41% higher.

We can conclude that if the best sequence (S_1) is selected, the impacts of the film with PVC and EVOH are similar. On the other hand, if other sequences with higher impacts are selected, the potential savings for recycling the material via STRAP™ could be significantly reduced. This analysis shows the applicability of this fast computational framework to guide the product design of easy-to-recycle multilayer plastic film or guide plastic waste sorting by identifying plastic wastes that can be sorted together, reducing the time and cost of sorting plastics (i.e., polymers) into individual bails.

3.4. Case Study 4

This last case study aims to investigate the separation difficulty in polymer mixtures. To achieve this, we applied the first three steps of our computational framework and conducted a systematic solvent selection test for various polymer combinations based on our solubility database. Given a polymer mixture and a solvent library, the number of feasible separation sequences and potential solvent candidates can represent the difficulty of separating the polymers. Smaller values of these two numbers indicate a higher difficulty of separation. Note

that difficulty here refers only to the number of feasible sequences (i.e., the number of ways to separate the polymer mixture) and the ease of selecting solvent candidates for their separation. Thus, the last two steps of the framework, which include performing the process simulation, TEA, LCA, and ranking of impacts, were not included (see Fig. 3). This case study considers ten polymers - LDPE, high-density polyethylene (HDPE), EVOH, PP, polystyrene (PS), PC, PVC, PET, nylon 6 (N6), and nylon 66 (N66) - and the 60-solvent library, as previously mentioned.

We first study all possible binary combinations between these ten polymers. We find that there are 45 unique binary combinations. Every binary mixture has a maximum of two possible separation sequences. For instance, in a mixture of LDPE and PC, one sequence refers to dissolving LDPE and recovering PC via filtration. Thus, the other sequence involves dissolving PC and recovering LDPE via filtration. We identify the solvent candidates available in our database for both sequences. If no solvents are available for a given sequence, we consider that sequence to be unfeasible. The solvent selection process follows the assumptions and steps described in Section 2.

Fig. 9 presents a visualization of the difficulty for selective separation of all possible binary mixtures comprising the selected ten polymers. Here, we represent the difficulty (easy, medium, hard, and

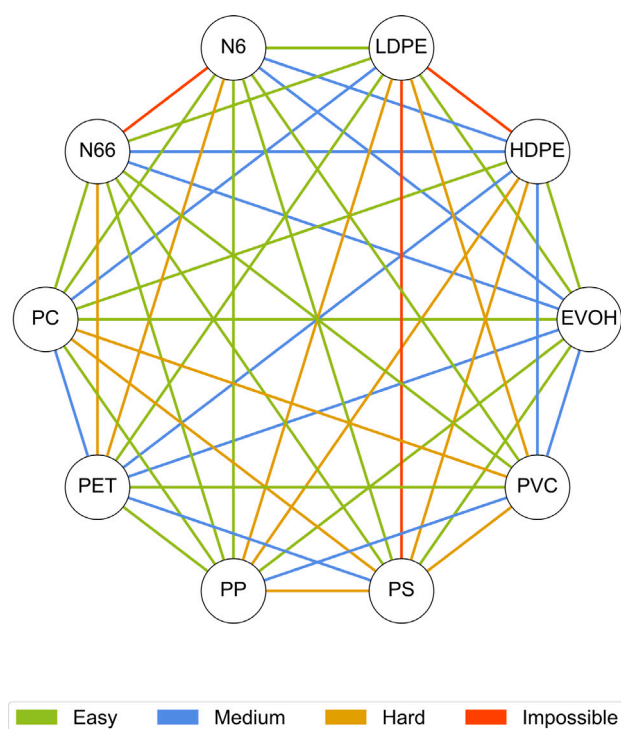


Fig. 9. Separation difficulty in binary polymer mixtures. Difficulty refers to the number of feasible separation sequences and potential solvent candidates.

impossible) by lines of different colors (green, blue, yellow, and red) connecting each binary mixture. We determine the difficulty for each binary polymer combination as follows. We sum the number of solvent candidates with the number of feasible separation sequences (which can be one or two for a binary mixture), and we establish a set of ranking criteria based on this value. If the value is >10 , we rank the separation as “easy”, but if the value is between 5 and 10, we rank it as “medium”. For the mixtures that have a value <5 , they are classified as “hard”, while “impossible” means that the value is 0. We specify these constraints based on the observations from the previous case studies. Note that the impossible case here means no solvents can be found to separate the binary polymer combination under the current conditions. The polymer combinations with impossible ratings may still be separable when a larger search space is considered (e.g., more solvents, larger temperature range, other precipitation methods, solvent mixtures). For example, in the case of the PET/PC combination, only PET can be dissolved (we did not find solvents that can dissolve PC), which means that the number of separation sequences is 1 since PET is dissolved first. The number of candidate solvents for selective PET dissolution in the presence of PC is 9. If we consider the sum of these values, we can see that PET and PC separation has a ranking of 10, which is considered “medium” difficulty (see Fig. 9).

From Fig. 9, we can see that, among all 45 combinations, there are 20 easy, 12 medium, 10 hard, and 3 impossible ratings. The hard and impossible combinations typically comprise two similar polymers: LDPE/PP, LDPE/HDPE, LDPE/PS, N6/N66. This result aligns with the general chemical idea of “like dissolves like”, as finding a solvent with very different solubility values for similar polymers is challenging. The ratings and the data here not only guide the design of plastic recycling processes but can also help plastic manufacturers design more sustainable and easy-to-recycle products or discover new applications of such inseparable polymers to eliminate the need for their separation.

As a complement to binary combinations, we also conducted similar calculations on combinations of more than two polymers. The calculation results (number of feasible separation sequences and solvent

candidates) for 3-polymer to 6-polymer combinations are available in the **Supporting Information**. As mentioned in Section 2.2.1, we consider the possibility of dissolving two polymers or more in a certain step. Note that several sequences can include the same solvents when these combinations have more than two polymers. The number of solvent candidates here has already excluded such duplicates. An obvious observation of these calculation results is that all impossible polymer combinations here are supersets of an impossible binary combination. Similarly, hard combinations of multiple polymers are often supersets of hard binary combinations. In short, these data can serve as supplementary reference for the design of multicomponent plastic waste recycling processes.

4. Conclusions and future work

This study proposed a computational framework that integrates molecular-scale models, process modeling, TEA, and LCA to provide insights into the design of sustainable solvent-based recycling processes and easy-to-recycle multilayer/mixed plastics. The framework provides a fast estimation of the economic and environmental impacts of different process design scenarios for hard-to-recycle plastics (e.g., multilayer films). The framework reports the economic impact as the minimum selling price of the recovered products, while the environmental impact is evaluated as the climate change impact. Each scenario includes different separation sequences, solvents, and process operating conditions (polymer/solvent ratio, dissolution temperature, solvent recycling rate). We also provide guidelines for product design by identifying multilayer film designs with a lower recycling impact. The computational framework is general and can be used for complex multilayer films or multicomponent plastic waste streams.

We focused on a process denominated as the STRAPTM technology, which can selectively dissolve and separate the constituent polymers of multicomponent plastics. We addressed 4 case studies to illustrate the applicability of the proposed framework. For Case Study 1, we analyzed the impacts of recycling a multilayer film composed of LDPE, EVOH, PET, and N6. We identified the sequence and process designs that drive the impacts. We found that recovering the component with higher weight composition (LDPE) first led to lower impacts. In addition, when other components were recovered first, the impacts could only be reduced if all the polymers were recovered. Overall, dissolving and recovering LDPE had lower impacts, while dissolving EVOH or recovering N6 via filtration led to higher impacts. For Case Study 2, we evaluated the feasibility and the economic and environmental impacts of recycling a more complex 7-polymer feedstock with LDPE, PET, PP, PC, EVOH, PVC, and N6. Here, we identified 36 feasible sequences and presented the process configurations with the lowest impacts. We found that, in this mixture, recovering the component with the highest weight composition first did not provide the lowest impact. Surprisingly, such sequences that targeted the highest component by weight fraction were not feasible. We also found that the solvents and conditions required to recover LDPE in a 4-component film change from the solvents and conditions to recover LDPE in a 7-component mixture. We conclude that for this more complex mixture, the solvent selection has a greater influence on the impacts than the percentage composition. For Case Study 3, we analyzed how the impacts of the STRAPTM process changed with different multilayer film designs. We compare a film composed of LDPE, PVC, and N6 with a film containing LDPE, EVOH (instead of PVC), and N6. We presented the configurations with the lowest costs and emissions for both materials. We found that if the best sequence was selected, the impacts of the film with PVC and EVOH were similar. However, if other sequences with higher impacts were chosen, the material containing EVOH could lead to higher impacts. Finally, for case study 4, we ranked the difficulty of separating all possible binary polymer mixtures in a set of 10 common polymers (LDPE, HDPE, EVOH, PP, PS, PC, PVC, PET, N6, and N66). We considered the number of feasible separation sequences and candidate solvents for ranking the

separation from “easy” to “impossible”. These findings can provide insights into recycling a given material before applying the complete computational framework.

We considered specific assumptions, including the 60-solvent database, fixed processing capacity, and temperature-controlled precipitation. Future work could include varying these parameters and considering more solvents and other types of multicomponent plastic feedstocks. As mentioned above, other dissolution methods, such as antisolvent addition, could lead to higher economic and environmental impacts. However, such methods may be required for certain feedstocks or polymers. Therefore, an analysis that proposes the optimal combination of dissolution methods could provide a broader set of possibilities for recycling diverse multicomponent plastics. In addition, the proposed computational framework could be used to generate tailored datasets that can help predict the impacts of different feedstocks. Apart from the separation of polymers, another challenge in the recovery of high-quality polymer resins from plastic waste is the removal of contaminants. These contaminants in recycled polymers are often plastic additives that are used to enhance the performance of plastic products, such as plasticizers, stabilizers, flame retardants, and pigments. Other contaminants also find their way into waste plastics via material recovery facilities or through exposure to the environment. This proposed computational framework can be extended to include solvent selection and process design for contaminant removal steps in a plastic recycling process. As many plastic additives are typically mixtures of large families of chemical species, future studies on the clustering of these contaminants and property prediction models will be needed to support the process design framework.

CRedit authorship contribution statement

Ugochukwu M. Ikegwu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Panzheng Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Reid C. Van Lehn:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Victor M. Zavala:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Aurora del Carmen Munguía-López:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compchemeng.2025.109148>.

Data availability

Data will be made available on request.

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